

numerous generation tasks, including the synthesis of images, speech, 3D shapes, and graphs.

The Latent Diffusion architecture, implemented in Stable Diffusion, comprises of two steps:

1. **Forward Diffusion** - Gradually perturbs the input data by mapping it to noise using a simple stochastic process, achieved through a Gaussian diffusion kernel. This step is used only during training and not during inference.
2. **Parametrized Reverse** - Reverses the forward diffusion process and performs iterative denoising, generating realistic data by converting random noise. This step represents data synthesis and is trained to generate data.

Since both processes require sequential repetition of thousands of steps, injecting and reducing noise, the entire process is computationally expensive. To mitigate this, Stable Diffusion implements the diffusion process over a lower dimensional latent space rather than the actual pixel space. For instance, the autoencoder in Stable Diffusion uses a reduction factor of 8, which reduces the image shape from (3, 512, 512) to (3, 64, 64) in latent space, thereby reducing memory usage by 64 times. This makes it possible to train on limited resources while retaining flexibility and quality.

Stable Diffusion Architecture

The main components of the Stable Diffusion architecture are designed to reduce samples to a lower-dimensional latent space and then denoise random

